

433.4MHz by default. The maximum transmitting power of the module is 100mW (20dBm). The data sheet indicates that it can communicate up to 1000 metres at 1200 baud rate, and 600 metres at 9600 baud rate (by default), when both transmitter and receiver are placed without any obstruction in between and are visible to each other.

The built-in MCU performs the communication with external device and no programming or configuration is required for basic use of the module. To change any parameters, its SET pin has to be pulled low and AT commands used. By default, the on-board pull-up resistance makes the SET pin high and transmits and receives the wireless data (even without any connection to SET pin).

DS1307 is a popular real time clock

(RTC) chip with built-in date and time format and calendar. It uses a 32.768kHz crystal to generate an accurate clock pulse and communicates using a two-wire interface (TWI, also called I2C) with the microcontroller. It has to be powered with 3V battery permanently to run the date and time internally. A 5V DC power supply is required to read or write the date and time from the chip.

Initial setup

After assembling the circuit on PCB, insert the microcontroller, HC-12 module, and the RTC IC in their respective places. Put the 3V battery in its holder and connect LCD1. Use SJ1, SJ2, and SJ3 shorting jumpers to set an ID number for the board. Switch on power using switch S1 after connecting 9V or 12V DC power supply. Power LED1, seconds LED4, and back-light of LCD1 will start glowing.

Now, write/burn *GroupMessenger.hex* file into the microcontroller using a suitable AVR programmer through ISP port. Once the program is written to the microcontroller, disconnect the programmer. Use shorting jumpers (refer table) to set ID number for each board. This has to be done only once on initial setup and then left alone.

The program goes through self-test and checks the connections to the external modules. The sequence of

testing is as follows with the blinker LED2 blinking for checking every component:

1. LCD1 shows 'Group Messenger' message for a while.

2. The program checks internal EEPROM memory and writes the default usernames in case of new microcontroller. Then reads the saved names from the EEPROM.

3. Checks and displays the ID number (1 to 8) and name according to the SJ1 to SJ3 jumper settings.

4. The new-message LED3 glows for a while.

5. A short beep is heard through the piezo buzzer (PZ1).

6. Sets USART for 9600 baud rate and displays the same on LCD1.

7. Checks the connections to HC-12 module and displays the Ok/Error message.

8. Checks the status of RTC chip and resets to default values, if required. The seconds LED4 starts blinking now, else shows Error message if connections are not proper.

9. Enters the main program loop and displays the date and time continuously.

To change your ID name or date and time on first-time use, press ES5 button (Setting). The current username is displayed on LCD1 and cursor blinks for editing the name. Use ES1 (move cursor to next position), ES2 (move cursor to previous position), ES3 (for next ASCII character), and ES4 (for previous ASCII character) for editing the required username. If no button is pressed for more than five seconds, the edited/new username gets saved to EEPROM of the microcontroller. Then the current date and time is displayed on LCD1 along with cursor. Use ES1 (move cursor to next position) and ES3 (increment

Status of Shoring Jumpers

Table 1

SJ1	SJ2	SJ3	User ID Number
SHORT	SHORT	SHORT	1
OPEN	SHORT	SHORT	2
SHORT	OPEN	SHORT	3
OPEN	OPEN	SHORT	4
SHORT	SHORT	OPEN	5
OPEN	SHORT	OPEN	6
SHORT	OPEN	OPEN	7
OPEN	OPEN	OPEN	8

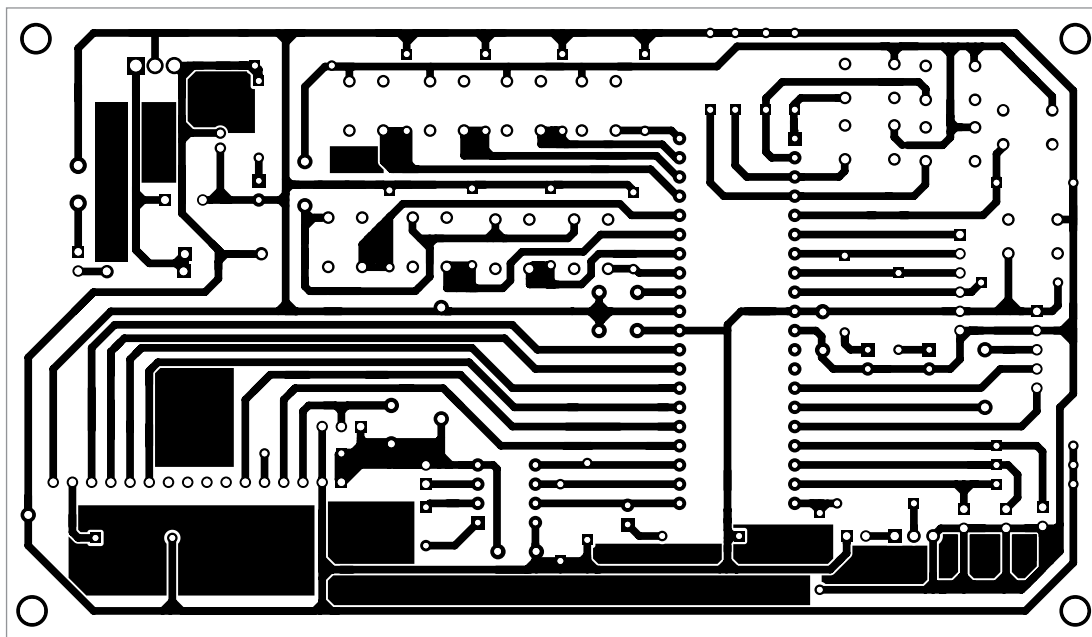


Fig. 2: Actual-size PCB layout for group messenger